

“U.G.” END SEMESTER EXAMINATION – JANUARY, 2025
1ST YEAR (1ST SEMESTER) – MECHANICAL ENGINEERING
ENGINEERING MECHANICS
MENUGES01

[Full Marks: 80]

[Time: 3:00 hrs.]

Assume suitable data, if not provided

[15X1 = 15]

A. Multiple choice questions (Compulsory) –

1. If the resulting moment about the door pivot axis at O is 35 N-m (figure A1), the force magnitude is –

a) 16.65 N b) 22.65 N c) 32.65 N d) None of these

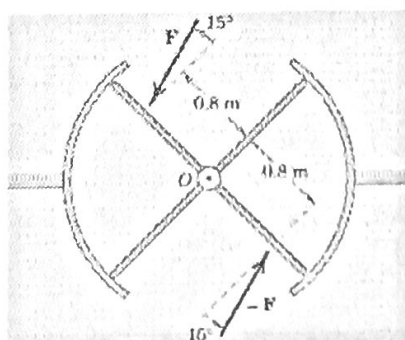


Figure A1

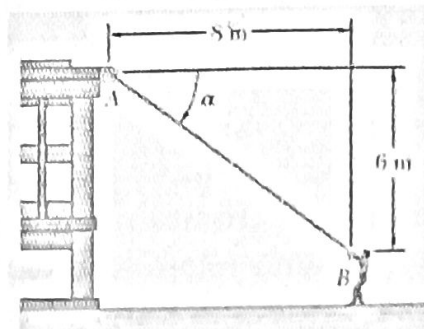


Figure A2

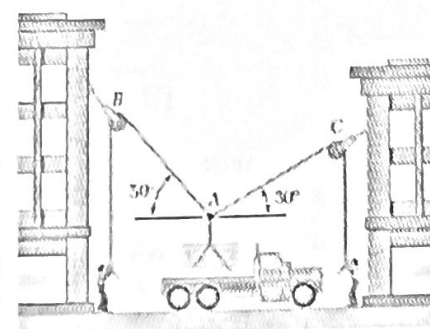


Figure A3

2. A man pulls with a force of 300 N on a rope attached to the top of a building, as shown in figure A2. This force can be written vectorially as –

a) $\vec{F} = -240\hat{i} + 180\hat{j}$ N b) $\vec{F} = 240\hat{i} - 180\hat{j}$ N
 c) $\vec{F} = 240\hat{i} + 180\hat{j}$ N d) $\vec{F} = -240\hat{i} - 180\hat{j}$ N

3. The values of tension ropes T_{AB} and T_{AC} under equilibrium conditions (figure A3) become –

a) $T_{AB} = 547$ N and $T_{AC} = 480$ N b) $T_{AB} = 647$ N and $T_{AC} = 380$ N
 c) $T_{AB} = 647$ N and $T_{AC} = 480$ N d) $T_{AB} = 547$ N and $T_{AC} = 380$ N

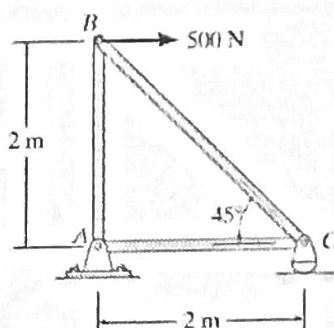


Figure A4

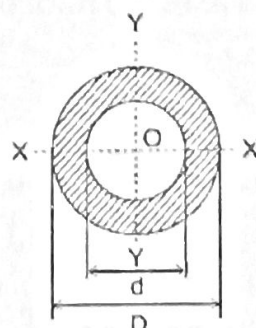


Figure A5

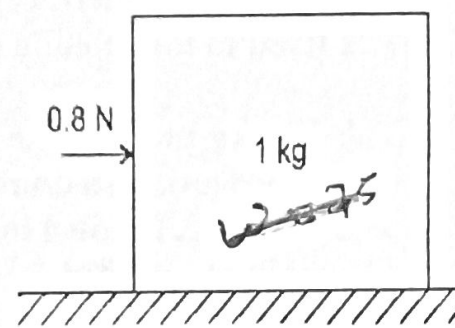


Figure A6

4. In figure A4, the member AC is in –
 a) Tension b) Compression c) Neutral d) None of these
5. The moment of inertia in a hollow section shown in figure A5 with respect to X-X' for 20 mm external and 10 mm inner diameters becomes –
 a) 8469.72 mm⁴ b) 6472.15 mm⁴ c) 9363.10 mm⁴ d) 7363.10 mm⁴
6. A 1-kg block rests on a surface with its static and dynamic coefficient of friction of 0.31 and 0.21, respectively. A 0.8 N force is applied to the block, as shown in figure A6. The friction force becomes –
 a) 2N b) 3N c) 4N d) 5N
7. A smooth cylinder lying on a _____ surface is in neutral equilibrium.
 a) Concave b) Convex c) Horizontal d) Zig-Zag

8. A thin disc and a thin ring, both have mass M and radius R . Both rotate about axes through their center of mass and are perpendicular to their surfaces at the same angular velocity. Which of the following is true?
- The ring has higher kinetic energy
 - The disc has higher kinetic energy
 - The ring and the disc have the same kinetic energy
 - Kinetic energies of both the bodies are zero, since they are not in linear motion
9. If two forces each equal to P in magnitude act at right angles, their effect may be neutralized by a third force acting along their bisector in opposite direction whose magnitude is –
- $2P$
 - $0.5P$
 - $0.707P$
 - $1.414P$
10. A framed structure with joints j is said to be perfect, if it contains members equal to –
- $2j - 3$
 - $3j - 2$
 - $2j - 1$
 - $j - 2$
11. Forces $5i + 6j + 7k$ N and $-8i - 9k$ N are acting at a point. The resultant force will be of magnitude –
- 5 N
 - 7 N
 - 9 N
 - 11 N
12. A body is said to be in stable equilibrium state, if the second derivative of its total potential energy is _____ zero.
- equal to
 - greater than
 - less than
 - none of these
13. The position coordinate of a particle which is confined to move along a straight line is given by $s = t^3 - 12t + 6$, where s is measured in meters from a convenient origin and t is in seconds. The acceleration of the particle at $t = 4$ s is –
- 24 m/s^2
 - 48 m/s^2
 - 66 m/s^2
 - 88 m/s^2
14. If the expression of linear momentum is $\vec{G} = \frac{3}{2}(t^2 + 3)\hat{j} - \frac{3}{4}(t^4 - 4)\hat{k}$, then the force in Newton will be –
- $t\hat{j} - 2t^2\hat{k}$
 - $2t\hat{j} - 2t^3\hat{k}$
 - $3t\hat{j} - 3t^3\hat{k}$
 - $3t\hat{j} - 2t^3\hat{k}$
15. Four particles A, B, C and D of mass $\frac{m}{2}$, m , $2m$ and $4m$ have the same momentum, respectively. The particle with maximum kinetic energy is –
- A
 - B
 - C
 - D

B. Short questions (Answer any five questions):

[5X5 = 25]

1. The frame shown in **figure B1** supports part of the roof of a small building. Knowing that the tension in the cable is 150 kN, determine the reaction at the fixed end E. [5]

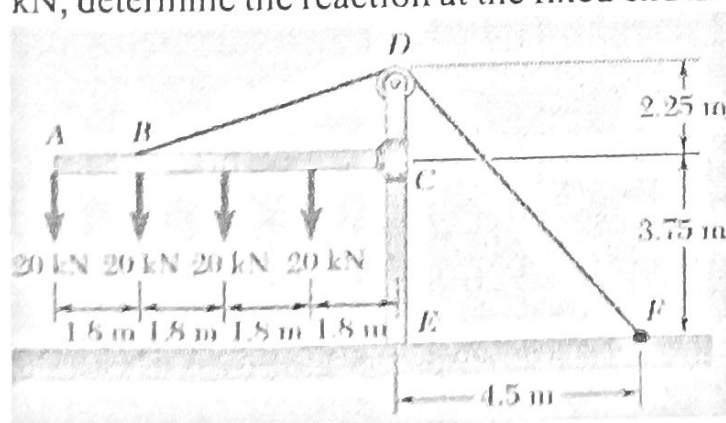


Figure B1

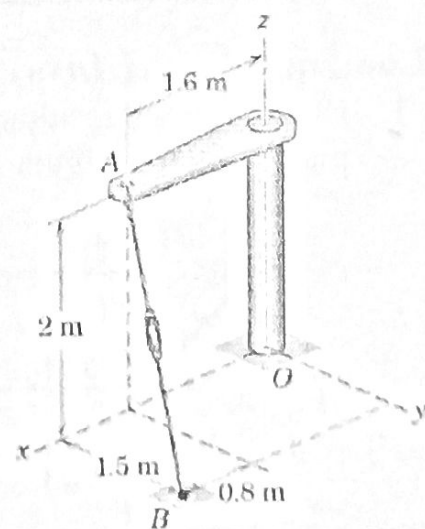


Figure B2

2. The turnbuckle is tightened until the tension in cable AB is 3.4 kN. Determine the moment about point O of the cable force acting on point A (figure B2). [5]

3. Compute the volume V and mass of the solid generated (material is steel) by revolving the right triangle about the z -axis (figure B3) through 180° . (Consider the density of steel is 7830 kg/m^3) [3+2=5]

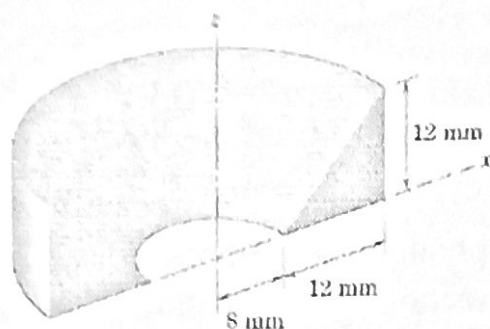


Figure B3

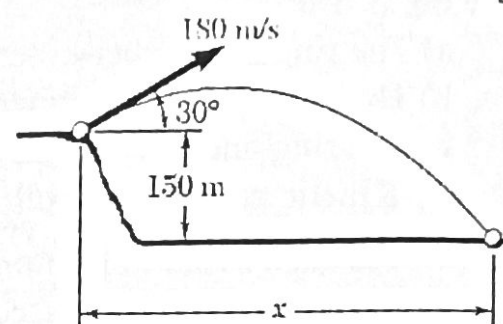


Figure B4

4. A projectile is fired from the edge of a 150 m cliff with an initial velocity of 180 m/s at an angle 30° with the horizontal as shown in figure B4. Neglecting air resistance, find the horizontal distance from the gun to the point where the projectile strikes on the ground. [5]
5. A 75 kg man stands on a spring scale in an elevator. During the first 3 seconds of motion from rest, the tension T in the hoisting cable is 8300 N. Find the reading R of the scale in kg during this interval. (no figure) [5]
6. A 120-gm baseball is pitched with a velocity of 24 m/s toward a batter. After the ball is hit by the bat B, it has a velocity of 36 m/s in the direction shown in figure B6. If the bat and ball are in contact for 0.015 s, determine the average impulsive force (magnitude and direction) exerted on the ball during the impact. [5]

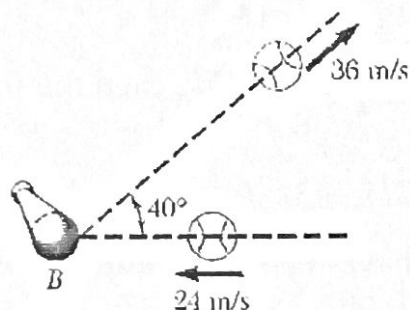


Figure B6

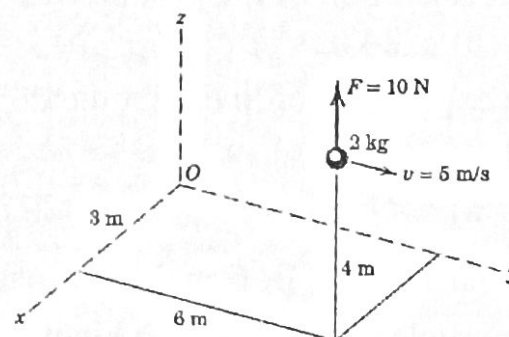


Figure B7

7. A small sphere has the position and velocity indicated in the figure B7 and is acted upon by the force F . Determine the angular momentum H_O about point O and the time derivative $\dot{H}_O$. [3+2=5]

C. Long questions (Answer any four questions):

[4X10 = 40]

1. Determine the resultant (force and couple) of the four forces and one couple which act on the plate shown in figure C1 [5+5=10]

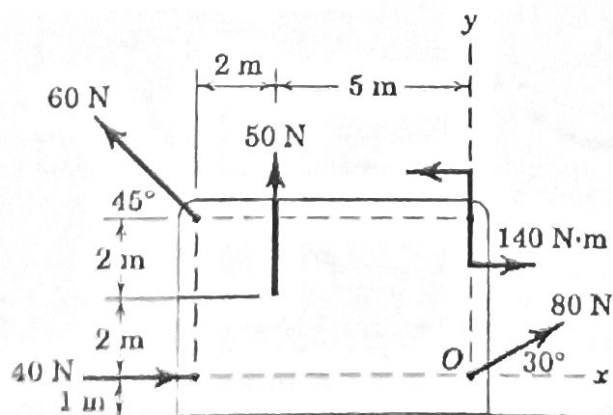


Figure C1

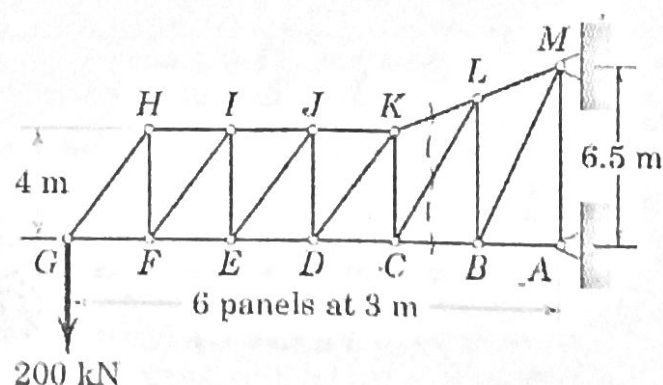


Figure C2

2. Calculate the forces induced in members KL, CL, and CB by the 200-N load on the cantilever truss (figure C2). [3+3+4=10]
3. Each of the two uniform hinged bars has mass m and length l , and is supported and loaded as shown in figure C3. For a given force P , determine the angle θ for equilibrium. Find the angle generated by the system under 50 N applied force. (Consider the mass of each bar is 5 kg of length 0.5m. [8+2=10]

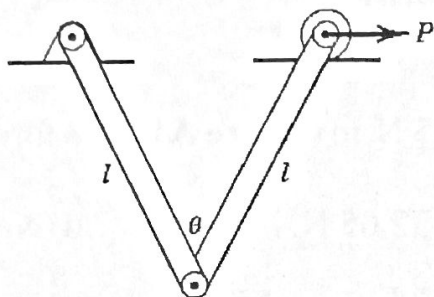


Figure C3

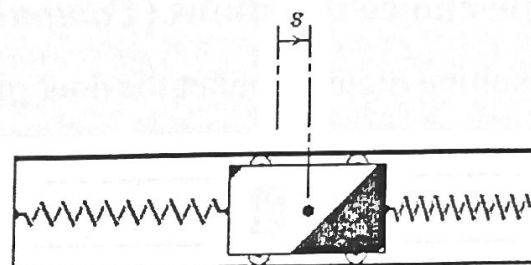


Figure C4

4. The spring-mounted slider moves in the horizontal guide with negligible friction and has a velocity v_0 in the s -direction as it crosses the mid-position, where $s = 0$ and $t = 0$ (figure C4). The two springs together exert a retarding force to the motion of the slider, which gives it an acceleration proportional to the displacement but oppositely directed, i.e., $a = -k^2s$, where $k = \text{constant}$. Determine the expressions for the displacement s and velocity v as functions of time t . [5+5=10]
5. Aircraft B has a constant speed of 150 m/s as it passes the bottom of a circular loop of 400-m radius as shown in figure C5. Aircraft A flying horizontally in the plane of the loop passes 100 m directly below B at a constant speed of 100 m/s. Determine the instantaneous velocity and acceleration which A appears to have to the pilot of B, who is fixed to his rotating aircraft. [4+6=10]

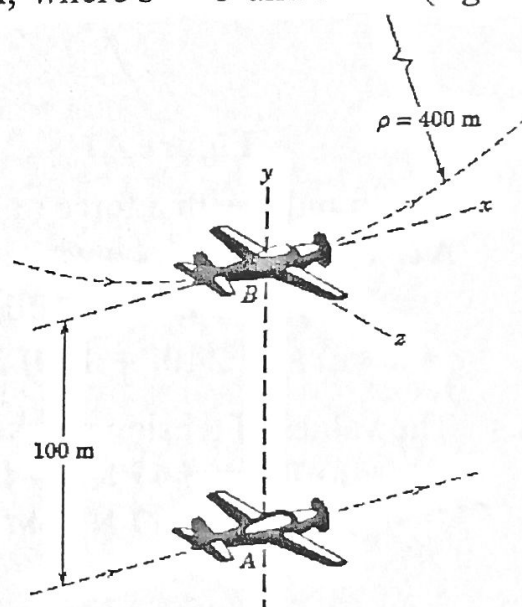


Figure C5

6. A 120-kg block is suspended from an inextensible cable that is wrapped around a drum with a 0.4-m radius that is rigidly attached to a flywheel shown in figure C6. The drum and flywheel have a combined centroidal moment of inertia of $\bar{I} = 16 \text{ kg} \cdot \text{m}^2$. At the instant shown, the velocity of the block is 2 m/s directed downward. Knowing that the bearing at A is poorly lubricated so that the bearing friction is equivalent to a couple M of magnitude 90 N.m, determine the velocity of the block after it has moved 1.25 m downward. [10]

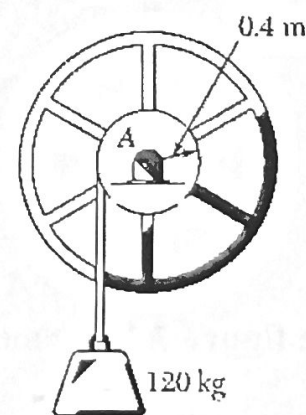


Figure C6

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